🎯 CareerGap AI — Skill Gap Analysis & Career Roadmap

Know Your Skill Gap. Build Your Career.

CareerGap AI is a web-based career guidance application designed to help students identify the skills they need to learn for their target job role.

Many B.Tech and CSE students know the job they want but are unsure about the technical skills, tools, projects, and certifications required to become job-ready. CareerGap AI addresses this problem by analyzing a student's current skills against the requirements of their selected career path and providing a personalized learning roadmap.

🌐 Live Application

Live Demo:
https://attractive-career-path-flow.base44.app

📌 Problem Statement

Students often struggle to understand:

- Which skills are required for their target job?
- Which skills do they already have?
- Which skills are missing?
- Which skills should they learn first?
- Which projects can help them improve their skills?
- How prepared are they for their target career?

Existing career resources can provide large amounts of information, but students may still find it difficult to convert that information into a clear and personalized learning plan.

Proposed Solution

CareerGap AI provides a personalized skill-gap analysis based on:

- Student's education
- Target job role
- Current technical skills
- Soft skills

- Projects
- Certifications
- Experience

The application identifies skill gaps and generates a structured roadmap to help students become more job-ready.

🎯 Objectives

The main objectives of CareerGap AI are:

1. Help students identify the skills required for their target career.
2. Compare current skills with career-specific requirements.
3. Identify missing and recommended skills.
4. Prioritize skills based on importance.
5. Generate a personalized learning roadmap.
6. Recommend suitable projects.
7. Help students track their learning progress.
8. Improve career awareness among college students.

💡 Key Features

👤 Student Profile

Students can create a career profile containing:

- Name
- Degree
- Branch
- Year of study
- Target job role
- Technical skills
- Soft skills
- Projects

- Certifications
- Experience

🔍 Skill Gap Analysis

The application compares the student's current skills with the skills required for their selected job role.

Example:

Skill	Status	Priority
Python	✅ Already Have	—
HTML/CSS	✅ Already Have	—
SQL	❌ Missing	High
Django	❌ Missing	High
REST API	❌ Missing	High
Git/GitHub	⚠️ Improve	Medium

📊 Career Readiness Score

The application provides an overall career-readiness score based on the student's current profile and required skills.

Example:

Career Readiness

 72%

Technical Skills 75%

Projects 60%

Certifications 70%

Soft Skills 80%

🗺️ Personalized Learning Roadmap

CareerGap AI generates a structured roadmap.

Example:

Phase 1 — Foundation

↓

Learn SQL

↓

Learn Git & GitHub

Phase 2 — Backend

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Learn REST APIs

↓

Learn Django

Phase 3 — Projects

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Build a REST API

↓

Build a Full Stack Project

Phase 4 — Job Preparation

↓

Resume Preparation

↓

Interview Preparation

💼 **Career Explorer**

Students can explore different career paths such as:

- Web Developer
- Frontend Developer
- Backend Developer

- Full Stack Developer
- Python Developer
- Java Developer
- Data Analyst
- Data Scientist
- AI/ML Engineer
- Cybersecurity Analyst
- Cloud Engineer

🚀 Project Recommendations

The application recommends projects based on the student's skill gaps.

For example:

Missing Skill:

REST APIs

Recommended Project:

Student Management REST API

Difficulty:

Intermediate

Skills Developed:

Python

REST API

Database

Backend Development

📈 Progress Tracking

Students can track their learning progress by marking skills as:

- Not Started

- Learning
- Completed

🤖 AI Integration

Artificial Intelligence is used to make the application personalized rather than simply displaying static career information.

The AI-assisted functionality includes:

- Skill-gap identification
- Career-readiness analysis
- Skill prioritization
- Personalized learning roadmap
- Project recommendations
- Career recommendations
- Next-skill suggestions

The application can continuously update recommendations based on the student's progress.

🛠️ Technology Stack

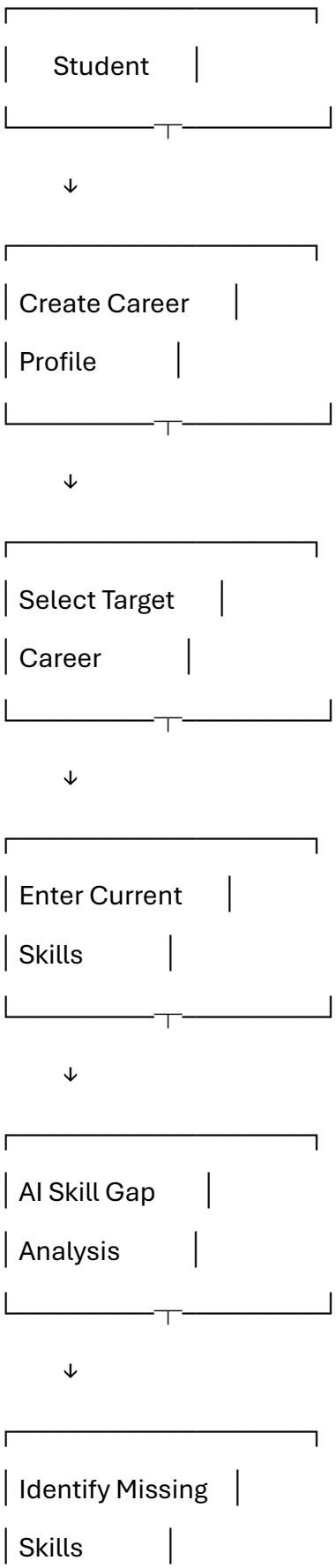
Application Development

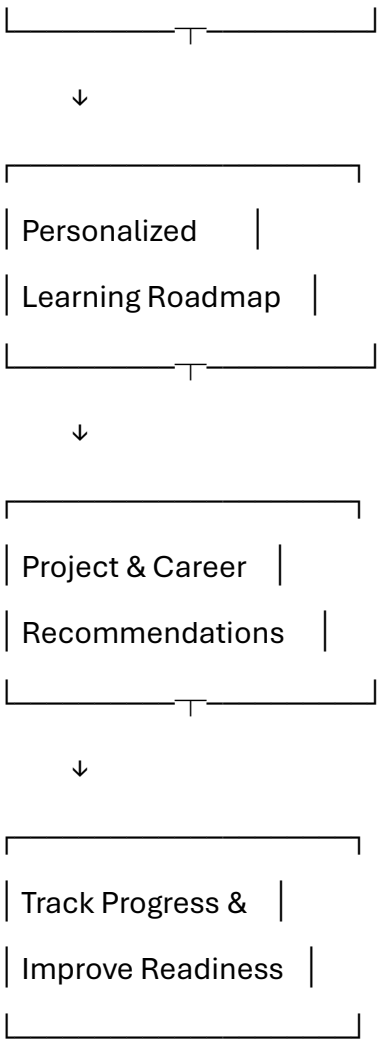
- **Base44 AI** — AI-powered application development platform
- **Web Application** — Responsive browser-based interface
- **Database** — Application data storage
- **AI** — Skill-gap analysis and personalized recommendations

Why Base44?

Base44 allows applications to be created from natural-language requirements and supports application interfaces, backend logic, data management, authentication, and deployment. This made it suitable for rapidly developing and testing the CareerGap AI mini project.

System Workflow





👨‍💻 User Flow

1. Open CareerGap AI.
2. Create an account.
3. Enter academic information.
4. Select a target job role.
5. Enter current skills.
6. Submit the career profile.
7. View the AI-generated skill-gap analysis.
8. Identify high-priority missing skills.
9. Follow the personalized learning roadmap.
10. Complete recommended projects.

11. Update skill progress.
12. Recheck career readiness.

📊 **Example Use Case**

Student Profile

Degree: B.Tech

Branch: CSE

Target Role: Python Developer

Current Skills:

Python

C

HTML

CSS

CareerGap AI Analysis

Skills You Have:

✓ Python

✓ HTML

✓ CSS

✓ C

Skills to Learn:

❌ SQL

❌ Git/GitHub

❌ REST APIs

❌ Django

❌ Testing

Recommended Roadmap

1. SQL

2. Git & GitHub

3. REST APIs

4. Django

5. Testing

6. Build Backend Projects

🗂️ Main Data Entities

The application can maintain information such as:

Users

User ID

Name

Email

Education

Branch

Year

Career Profile

User ID

Target Role

Technical Skills

Soft Skills

Projects

Certifications

Experience

Career Roles

Role Name

Description

Required Skills

Recommended Skills

Technologies

Skill Gap

User ID

Skill

Current Status

Priority

Progress

Learning Roadmap

User ID

Skill

Learning Phase

Priority

Status

Estimated Learning Time

🎓 **Advantages**

- Personalized career guidance
- Easy to use
- Student-focused
- Saves career research time
- Identifies missing skills
- Provides structured learning direction
- Encourages project-based learning
- Tracks student progress
- Can be expanded for placement preparation

⚠️ Limitations

- Career requirements can change over time.
- Skill recommendations depend on the quality of available career information.
- AI recommendations should be treated as guidance rather than guaranteed job predictions.
- The current project is developed as a mini-project/MVP.

🔮 Future Scope

Future versions could include:

- 📄 AI Resume Analyzer
- 💼 Job Recommendation System
- 🎤 AI Mock Interview
- 🧠 AI Coding Assessment
- 📚 Personalized Course Recommendations
- 🏆 Certification Recommendations
- 📊 Placement Readiness Dashboard
- 🔗 Job-board integration
- 📈 Industry skill-demand analysis
- 👥 Student mentor matching
- 📱 Mobile application

🌟 Expected Impact

CareerGap AI aims to help students move from:

"I don't know what to learn."

↓

"I know what skills I am missing."

↓

"I know what to learn first."

↓

"I have a clear career roadmap."

↓

"I am becoming job-ready."

The goal is to make career preparation more structured, personalized, and accessible for students.

🚀 Deployment

The application is deployed using Base44 and is available online.

Live Application:
https://attractive-career-path-flow.base44.app

Base44 provides hosting and deployment capabilities that allow the generated application to be published and shared online.

📸 Screenshots

Add screenshots of your application here.

Example:

screenshots/

├── home.png

├── dashboard.png

├── career-profile.png

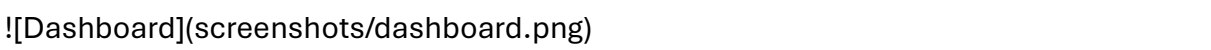
├── skill-gap.png

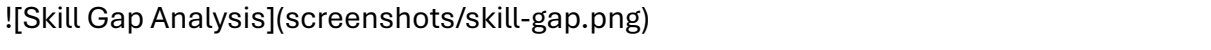
├── roadmap.png

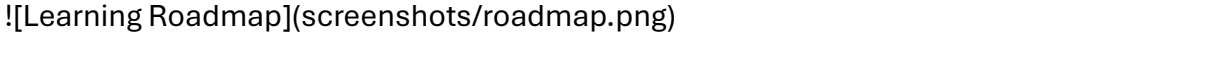
└── progress.png

Then add them to this README:

![Home Page](screenshots/home.png)







👷 Project Information

Project Name: CareerGap AI

Project Type: Mini Project

Domain: Artificial Intelligence / Career Guidance / Web Application

Target Users: B.Tech Students, CSE Students, Freshers

Development Platform: Base44 AI

Application Type: Web Application

📄 Conclusion

CareerGap AI provides a practical solution to a common problem faced by college students: understanding what they need to learn for their desired career.

By combining career information, skill-gap analysis, AI-powered recommendations, personalized learning roadmaps, project suggestions, and progress tracking, the application helps students create a clearer path toward becoming job-ready.

⭐ Support

If you find this project useful, consider giving the repository a ⭐ on GitHub.

Live Demo:
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